May Tiger

Software Engineer • Embedded & Audio Development

052-6771039 | may.tiger877@gmail.com | LinkedIn | GitHub | Website

SUMMARY

Software engineer with a strong foundation in low-level and embedded systems and hands-on experience delivering performance-critical solutions. I gravitate toward complex problems that reward clean architecture and continuous learning, and I bring the same rigor to audio software — building real-time DSP tools and instruments from first principles, driven by a love of music and sound.

EXPERIENCE

Linux Embedded Software Engineer 2022 – 2025

SolarEdge Technologies

- Core R&D engineer on SolarEdge's residential Smart Home platform — a Linux-based system coordinating multiple connected energy devices.
- Built and optimized backend services in C for the multi-device platform, prioritizing low latency and high throughput under real-time constraints.
- Engineered energy-management algorithms balancing household demand across multiple power sources for efficient utilization.
- Designed modular data structures enabling seamless integration of diverse devices over multiple communication protocols (e.g. MQTT).
- Built and validated hardware test environments and CMocka unit tests, collaborating with hardware and software teams across the full lifecycle. Stack: C, with Python and C#.

Audio Software Developer 2023 – Present

Independent

- Build real-time audio plugins, effects, and instruments in C++ with the JUCE framework, implementing DSP and MIDI from first principles.
- Engineer generative and physics-driven instruments with custom interfaces and fine-grained control over sound and timing.
- Apply machine learning to audio enhancement, including dereverberation and noise reduction, using PyTorch and Librosa.

Official Basketball Referee 2012–2014, 2017–2018

Israeli Basketball Association — officiated competitive games, making fast decisions under pressure.

EDUCATION

B.Sc. in Computer Science 2019 – 2024

Technion – Israel Institute of Technology

- **Focus areas:** Machine Learning, Deep Learning, Natural Language Processing, and Program Synthesis.
- **Core CS:** Data Structures, Algorithms, Operating Systems, and Compilation.

PROJECTS

BallPit — *physics-based MIDI generator (C++ / JUCE)*

- A MIDI generator plugin that harnesses the chaos and unpredictability of balls colliding in an enclosed space, offering extensive customization and a high degree of user control over both notation and rhythm.

Exodus Delay — *advanced audio delay effect (C++ / JUCE)*

- A real-time "individual-instance" delay effect that lets each feedback line be customized with its own effects and filtering, giving every delay tap a unique character.

Audio De-Reverberation Tool — *guided by Dr. Lior Arbel*

- Combined an LS-UNET with a UNET model to tackle reverberation artifacts and background noise; built custom deep-learning components from first principles and tuned via hyperparameter optimization, outperforming single-task baselines.

Optimal Running Pace via Bio-Feedback — *guided by Mr. Eran Zituni, Vista Lab*

- A personalized music-recommendation system that uses smartwatch biofeedback — real-time heart rate and cadence — to identify the running pace that best balances exertion and endurance.

MILITARY SERVICE

Combat Medic — *"Moran" Special Artillery Unit* 2014 – 2017

TECHNICAL SKILLS

Languages: C, C++, Python, Assembly, C# · **Audio & ML:** JUCE, real-time DSP, PyTorch, Librosa
Tools & Infrastructure: Linux, Git, GitHub Actions, CMake, CMocka, MQTT, Wireshark, Jira, Figma
Spoken Languages: Hebrew (native), English (native), Japanese (beginner)